

NIKOLAYEVSK-ON-AMUR, Russian Federation

WMO: 313690

Lat: 53.148N

Lon: 140.694E

Elev: 46

StdP: 100.78

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.5	(c) -30.7	(d) -35.3	(e) 0.1	(f) -32.2	(g) -33.4	(h) 0.2	(i) -30.3	(j) 13.5	(k) -13.0	(l) 11.0	(m) -14.9	(n) 1.3	(o) 350	(p) 0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.3	(c) 26.0	(d) 19.7	(e) 24.2	(f) 18.7	(g) 22.6	(h) 17.9	(i) 20.8	(j) 24.4	(k) 19.7	(l) 23.0	(m) 18.6	(n) 21.7	(o) 3.7	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.2	22.9	18.2	13.2	21.7	17.3	12.4	20.5	60.6	24.7	56.8	23.1	53.3	21.8	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.1	(b) 8.5	(c) 7.4	(e) -35.2	(f) 29.9	(g) 2.4	(h) 2.1	(i) -36.9	(j) 31.4	(k) -38.3	(l) 32.7	(m) -39.6	(n) 33.8	(o) -41.3	(p) 35.4		
(a) 10.1	(b) 8.5	(c) 7.4	(e) -35.3	(f) 22.7	(g) 2.3	(h) 1.6	(i) -37.0	(j) 23.9	(k) -38.4	(l) 24.8	(m) -39.7	(n) 25.7	(o) -41.4	(p) 26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-21.3	-18.9	-11.2	-2.1	5.5	13.0	16.8	16.5	11.0	2.3	-10.0	-19.2		
	DBStd	14.51	6.20	5.14	5.07	3.74	3.72	3.75	3.15	2.88	3.72	3.86	6.58	6.45		
	HDD10.0	4734	970	810	656	364	146	13	1	0	33	239	598	904		
	HDD18.3	7227	1228	1043	914	614	396	163	67	72	222	497	848	1162		
	CDD10.0	588	0	0	0	0	8	104	211	203	62	1	0	0		
	CDD18.3	39	0	0	0	0	0	4	19	16	1	0	0	0		
	CDH23.3	266	0	0	0	0	2	48	111	100	6	0	0	0		
	CDH26.7	47	0	0	0	0	0	11	24	13	0	0	0	0		
	WSAvg	3.3	3.0	2.8	3.3	3.9	4.2	3.7	3.5	2.9	2.8	3.3	3.3	3.4		
	PrecAvg	649	36	30	35	38	50	45	65	86	83	77	55	51		
	PrecMax	899	106	99	124	110	133	150	154	222	185	218	169	108		
(15) Precipitation	PrecMin	437	2	2	1	3	1	3	0	7	14	14	5	5		
	PrecStd	111	28	23	27	20	28	32	45	47	39	44	30	28		
	DB	-1.8	-2.9	3.0	9.9	20.8	28.1	29.9	28.5	24.1	15.9	5.0	-0.1			
	MCWB	-2.4	-3.9	-0.3	4.6	13.4	18.9	22.6	20.7	18.1	10.4	2.9	-0.5			
	2% DB	-4.1	-6.0	-0.1	6.8	17.2	24.1	26.0	25.9	21.8	12.4	2.1	-1.9			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	-4.6	-6.8	-1.9	3.0	11.3	17.5	19.8	19.9	16.5	8.8	1.0	-2.4			
	5% DB	-8.0	-8.6	-1.6	4.9	14.7	22.0	24.0	24.1	19.7	10.2	0.8	-4.6			
	MCWB	-8.6	-9.4	-2.9	1.7	9.8	16.3	18.9	19.1	15.4	7.4	-0.2	-5.1			
	10% DB	-12.1	-11.0	-3.0	3.1	12.2	20.1	22.2	22.4	17.5	8.2	-0.6	-9.5			
	MCWB	-12.5	-11.6	-4.4	0.7	8.4	15.4	18.2	18.3	14.2	6.1	-1.5	-9.9			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	-2.3	-3.6	0.1	5.2	14.0	20.2	22.7	22.5	19.3	11.5	3.2	-0.5			
	MCDB	-1.8	-3.0	2.1	8.7	20.1	25.7	28.0	26.2	22.6	14.1	4.5	-0.3			
	2% WB	-4.6	-6.9	-1.6	3.3	11.7	18.4	20.9	20.9	17.2	9.6	1.2	-2.5			
	MCDB	-4.1	-6.1	-0.4	6.2	16.2	22.9	24.7	24.3	20.5	11.8	1.7	-2.0			
	5% WB	-8.6	-9.5	-2.8	2.0	10.3	17.1	19.7	19.9	15.9	7.8	-0.1	-5.1			
(31) Mean Daily Temperature Range	MCDB	-8.1	-8.7	-1.6	4.1	13.8	21.0	22.9	22.9	18.7	9.6	0.5	-4.5			
	10% WB	-12.6	-11.7	-4.4	0.9	8.9	15.8	18.7	18.8	14.6	6.4	-1.6	-10.0			
	MCDB	-12.2	-11.1	-3.1	2.7	11.9	19.5	21.6	21.8	17.3	7.9	-0.7	-9.6			
	5% DB	7.4	9.5	10.7	8.6	8.8	10.0	8.3	9.1	10.2	8.1	7.6	6.9			
	MCDBR	9.0	9.5	10.6	11.2	13.9	14.1	12.4	12.8	13.5	11.8	7.3	7.0			
(36) Clear-Sky Solar Irradiance	MCWBR	8.9	8.7	8.8	8.2	8.8	8.4	7.3	7.6	8.5	8.3	6.4	6.9			
	5% WB	8.8	8.9	9.7	9.8	12.9	12.6	10.2	10.9	11.6	10.1	6.4	6.9			
	MCWBR	8.7	8.2	8.2	7.4	8.5	7.9	6.5	7.0	7.9	7.7	5.9	6.9			
	taub	0.286	0.312	0.384	0.493	0.555	0.522	0.523	0.457	0.391	0.393	0.344	0.299			
(40) All-Sky Solar Radiation	taud	2.000	1.964	1.807	1.642	1.715	1.932	1.981	2.184	2.311	2.105	2.014	1.960			
	Ebn at Noon	645	755	764	723	706	742	731	764	776	656	555	531			
	Edh at Noon	79	113	165	223	219	178	167	129	101	99	79	69			
(44) All-Sky Solar Radiation	RadAvg	1.05	2.15	3.50	4.52	4.89	5.42	4.82	4.10	3.19	1.80	0.97	0.72			
	RadStd	0.12	0.17	0.34	0.49	0.54	0.57	0.59	0.38	0.28	0.16	0.12	0.13			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-141	N/A	N/A
Regional (3 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-141	N/A	N/A

Nomenclature: See separate page